Week 10, Lecture 2: Bayesian Model Comparison and Shrinkage Priors

Instructor notes. Part 1: prep reading. Part 2: podium cheat sheet

Overview (instructor only)

Beat	Mode	Time
1. Big picture	BOARD	~3 min
2. Ridge is a prior	BOARD → R	~6 min
3. WAIC and LOO	BOARD → R	~12 min
4. Shrinkage priors	BOARD → R	~12 min
5. Discussion	BOARD	~5 min
If time: Bayes factors, RJMCMC	BOARD	

- No in-class WOD; section 2 is a short demo instead.
- R = Wednesday section of `CODE_Week10_ModelComparison.R` (needs Monday's section run first). Hand out the take-home WOD `WOD_Week10_Lecture2_PondModelsWithLOO.Rmd` at the end (due Monday).
- **Behind?** Skip the DIC row; in section 4 show only the interval plot and its pause.
- **Built before Monday was taught;** check what Monday and Tuesday covered and adjust.

Part 1: Lecture content

1. Big picture (~3 min) [BOARD]

- COMPARE (2), Bayesian side. Monday: penalties on count and size. Today: the same two ideas for MCMC fits
- No single $\hat{\theta} \Rightarrow$ no AIC; priors blur "number of parameters"

2. Ridge is a prior (~6 min) [BOARD → R]

- Setup: 8 survey routes, change in log abundance $y_i \sim \text{Normal}(\mu, \sigma = 0.15)$; μ = regional trend

The math, step by step (board).

1. *Monday's ridge fit* (one parameter, so one penalty term):

$$\text{PNLL}(\mu) = \underbrace{\sum_{i=1}^{n} \frac{(y_i - \mu)^2}{2\sigma^2}}_{\text{NLL}} + \lambda\mu^2$$

   (the NLL is the Normal density's exponent; the $\log\sqrt{2\pi\sigma^2}$ terms don't involve μ, so they drop)
2. *Bayes' rule*, then take $-\log$ (products become sums):

$$P(\mu \mid y) \propto \mathcal{L}(\mu)\, P(\mu) \quad \Rightarrow \quad -\log P(\mu \mid y) = -\log \mathcal{L}(\mu) - \log P(\mu) + \text{const.}$$

3. *Prior* $\mu \sim \text{Normal}(0, \tau)$, a skeptical prior centered on "no trend":

$$P(\mu) = \frac{1}{\sqrt{2\pi\tau^2}}\, e^{-\mu^2/(2\tau^2)} \quad \Rightarrow \quad -\log P(\mu) = \frac{\mu^2}{2\tau^2} + \text{const.}$$

4. *Put together:*

$$-\log P(\mu \mid y) = \sum_{i=1}^{n} \frac{(y_i - \mu)^2}{2\sigma^2} + \frac{\mu^2}{2\tau^2} + \text{const.}$$

5. *Match to step 1:* identical functions of μ when

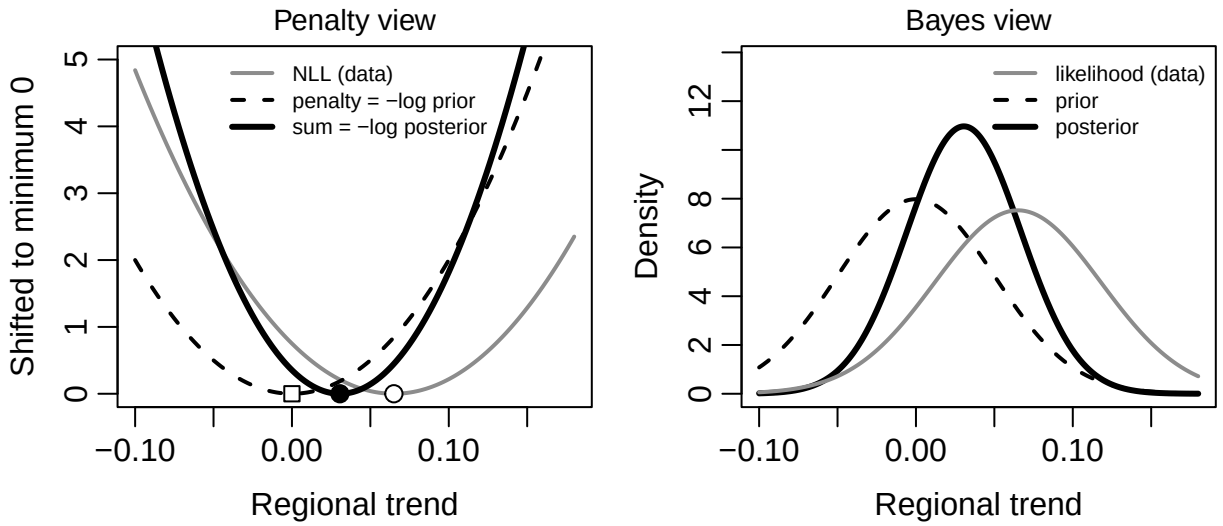
$$\lambda = \frac{1}{2\tau^2}$$

   so they have the same minimum: the **ridge estimate is the posterior mode**. Narrow prior (small τ) = big penalty; vague prior ($\tau \to \infty$) = $\lambda \to 0$ = MLE.
6. *Optional, the estimate itself:* set the derivative of step 4 to 0,

$$-\sum_{i=1}^{n} \frac{y_i - \mu}{\sigma^2} + \frac{\mu}{\tau^2} = 0 \quad \Rightarrow \quad \hat{\mu} = \frac{\sum y_i}{n + \sigma^2/\tau^2} = \bar{y} \cdot \frac{n}{n + \sigma^2/\tau^2}$$

   The MLE $\bar{y}$ times a shrinkage factor below 1. The prior acts like σ^2/τ^2 extra observations of exactly 0.
7. *Lasso:* the same matching with a Laplace prior, $P(\mu) = \frac{\lambda}{2} e^{-\lambda|\mu|}$, gives $-\log P(\mu) = \lambda|\mu| + \text{const.}$: the lasso penalty.

Numbers for the demo. $n = 8$, $\bar{y} = 0.065$, $\sigma = 0.15$, $\tau = 0.05$, so $\lambda = 200$ and $\sigma^2/\tau^2 = 9$ (the prior is worth 9 routes at 0). Shrinkage factor $8/17 = 0.47$, so $\hat{\mu} = 0.065 \times 0.47 = 0.031$. R: MLE 0.065; ridge 0.031 = posterior mode 0.031.



- Left: data's NLL (min at the MLE, open circle) + the prior's penalty (min at 0, square) = the ridge PNLL, which is the $-\log$ posterior (min in between, filled). Right: the same story as densities; the posterior sits between the likelihood and the prior, and is narrower than either (their precisions, n/σ^2 and $1/\tau^2$, add)
- **Pause:** which prior matches the lasso, $\lambda|\mu|$? *Land:* Laplace, density $\propto e^{-\lambda|\mu|}$ (spike at 0, heavier tails); section 4 uses it
- **Sticking point:** the mode is one point; the Bayesian payoff is the whole posterior (section 4)

3. WAIC and LOO (~12 min) [BOARD → R]

- Score = **elpd**: for each data point, log of its likelihood averaged over the posterior; sum. Higher is better; IC = $-2\times$ elpd
- Penalty = effective parameters, estimated from the posterior:
- DIC (Bolker): p_D = mean deviance − deviance at posterior means
- WAIC: p_{waic} = summed posterior variance of each point's log lik.
- LOO: approximates refitting without each point (PSIS); Pareto $k > 0.7$ flags points where the shortcut fails
- Stan must save `log_lik[i]` per observation (generated quantities)

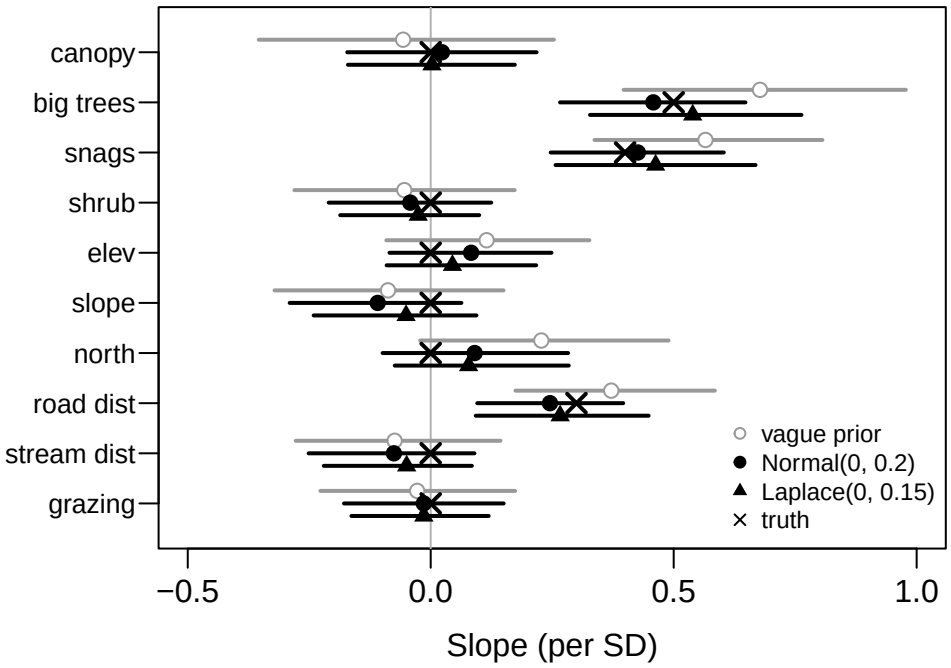
	no wave	full	effective parameters
AIC	2278.0	2282.0	k = 3 vs. 6
DIC	2278.0	2281.9	pD = 3.0 vs. 5.9
WAIC	2278.6	2282.7	p_waic = 3.5 vs. 6.5
LOOIC	2278.6	2282.7	p_loo = 3.5 vs. 6.5

- Fir: elpd difference -2.1 (SE 1.8); all Pareto k good
- **Pause:** full model 2.1 behind, SE 1.8. Conclusion? *Land:* about one SE; can't tell the models apart (not "no wave effect")

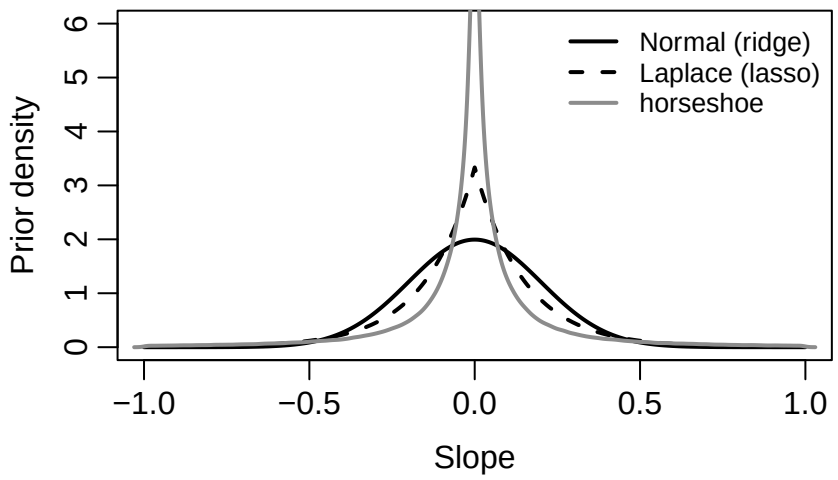
- **Sticking point:** AIC gives no SE for a difference; LOO does

4. Shrinkage priors (~12 min) [BOARD → R]

- Prior scale plays λ's role: Normal = ridge, Laplace = lasso
- Creepers (Monday), three priors on the 10 slopes:



- Northness (truth 0): vague 0.23 (-0.02 to 0.49); Laplace 0.08 (-0.07 to 0.28)
- p_{loo}: vague 9.1, Normal 7.8, Laplace 7.1 (10 slopes + intercept): tighter priors, fewer effective parameters
- LOO best: laplace; others within 1.5 elpd (SE up to 1.6). 3 stations have $k > 0.7$ (mention only)
- **Pause:** the lasso set northness to exactly 0; the Laplace posterior says 0.08 (-0.07 to 0.28). Which is more honest? *Land:* the posterior keeps the uncertainty; a Bayesian lasso never gives exact zeros
- Horseshoe: tall spike plus heavy tails (small effects shrink hard, big ones barely). Sketch only:



- **Sticking point:** choosing the prior scale = choosing λ; compare by LOO, or estimate it (a hierarchical prior, later)

5. Discussion (~5 min) [BOARD]

Q1. LOO prefers the no-wave fir model by 2.1 elpd (SE 1.8). A colleague writes "wave regeneration has no effect on cone production." Agree?

Q2. With a Laplace prior, northness gets a posterior median of 0.08 (90% interval -0.07 to 0.28); Monday's lasso said exactly 0. Which would you report, and why?

- Q1 lands on: a difference within ~1 SE is inconclusive; "not detected" $\neq$ "no effect"; corroborated, not proven
- Q2 lands on: the posterior shows the uncertainty a single zero hides; for prediction either works, for inference report the interval

If time: Bayes factors and selection inside the sampler [BOARD]

- Bayes factor = ratio of marginal likelihoods (averaged over the prior); sensitive to vague priors, hard to compute (Bolker 6.6.3)
- RJMCMC / spike-and-slab: the sampler jumps between models; posterior model probabilities. Needs discrete parameters, so not Stan (JAGS, NIMBLE); legacy LECTURE8 has BUGS code

PART 2: PODIUM CHEAT SHEET

Tight on purpose. Detail: Part 1. R = Wednesday section of CODE_Week10_ModelComparison.R.

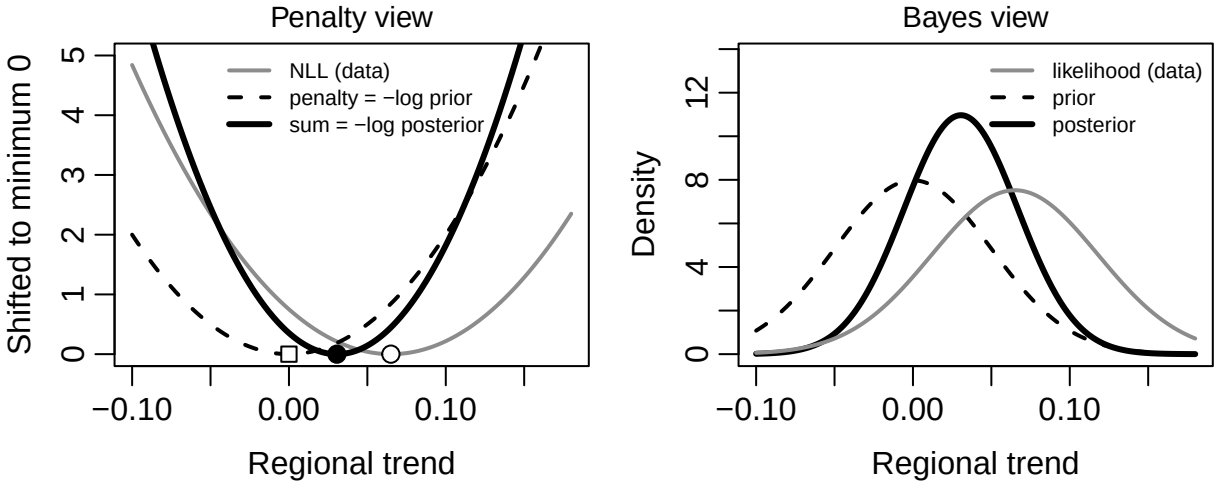
No in-class WOD; hand out the take-home LOO WOD at the end.
Behind? Skip the DIC row; in section 4 show only the interval plot and its pause.

1. Big picture (~3 min) [BOARD]

- Still COMPARE (2), now on the Bayesian side
- Monday: penalize the number of parameters (AIC) or their size (ridge, lasso). Today: both ideas for MCMC fits
- Why not just AIC? No single best-fit $\hat{\theta}$, and priors make "number of parameters" fuzzy

2. Ridge is a prior (~6 min) [BOARD → R]

- Ridge: $\text{PNLL}(\mu) = \Sigma(y_i - \mu)^2/(2\sigma^2) + \lambda\mu^2$
- Bayes, Normal(0, τ) prior: $-\log P(\mu \mid y) = \Sigma(y_i - \mu)^2/(2\sigma^2) + \mu^2/(2\tau^2)$ + const.
- Same function when $\lambda = 1/(2\tau^2)$: ridge estimate = posterior mode; vague prior ($\tau \to \infty$) = MLE
- Shrinkage: $\hat{\mu} = \bar{y} \cdot n/(n + \sigma^2/\tau^2)$; prior worth $\sigma^2/\tau^2 = 9$ routes at 0
- **R:** MLE 0.065; ridge 0.031 = posterior mode 0.031



- Left: NLL + penalty = $-\log$ posterior. Right: likelihood $\times$ prior $\propto$ posterior. The estimate lands between the data and 0
- **Pause:** "Which prior matches the lasso penalty, $\lambda|\mu|$?" *Land:* Laplace, $\propto e^{-\lambda|\mu|}$: a spike at 0 with heavier tails

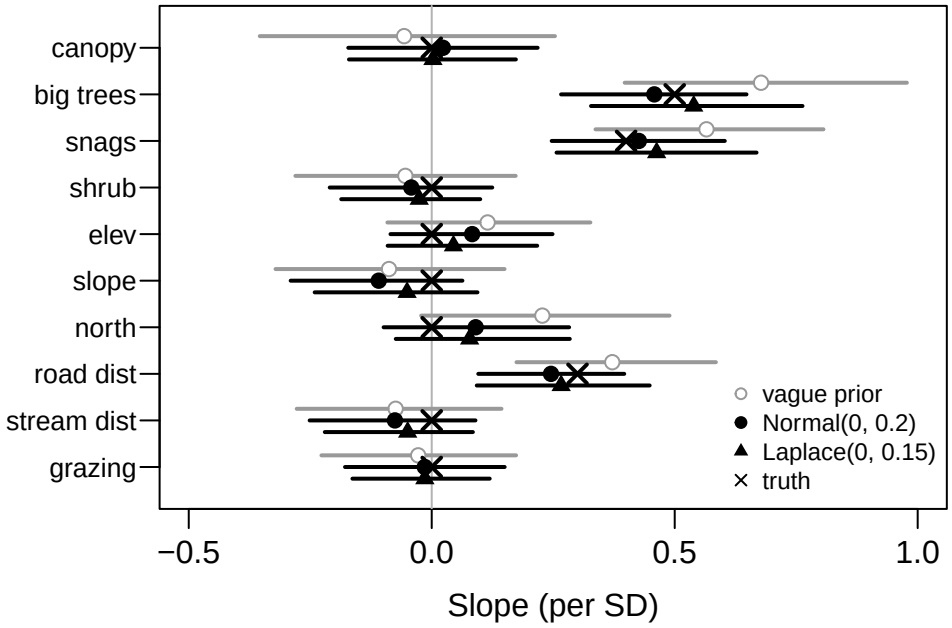
3. WAIC and LOO (~12 min) [BOARD → R]

- Score = elpd: for each point, log of its likelihood averaged over the posterior; sum. Higher is better; IC = $-2\times$ elpd
- Penalty = effective parameters from the posterior: p_D (DIC), p_{waic}, p_{loo}
- LOO approximates refitting without each point (PSIS); Pareto $k > 0.7$ flags points where the shortcut fails
- Stan must save `log_lik[i]` for every observation

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- **Pause:** "The full model is 2.1 elpd behind, SE 1.8. What can you conclude?" *Land:* about one SE: can't tell them apart (not "no wave effect")
- **Sticking point:** AIC gives no SE for a difference; LOO does

4. Shrinkage priors (~12 min) [BOARD → R]



- The prior's scale does λ's job: Normal prior = ridge, Laplace = lasso
- Northness (truly 0): vague prior 0.23; Laplace prior 0.08
- Effective parameters (p_{loo}) fall as the prior tightens: 9.1 → 7.8 → 7.1
- **Pause:** "Monday's lasso set northness to exactly 0. The Laplace posterior says 0.08 (90%: -0.07 to 0.28). Which is more honest?" *Land:* the posterior keeps the uncertainty a single zero hides; a Bayesian lasso never gives exact zeros
- Horseshoe (sketch only): a taller spike plus heavier tails; small effects shrink hard, big ones barely
- Choosing the prior scale = choosing λ: compare by LOO, or estimate it (hierarchical, later)

5. Discussion (~5 min)

Q1. LOO prefers the no-wave fir model by 2.1 elpd (SE 1.8). A colleague writes "wave regeneration has no effect on cone production." Agree?

- *Land:* a difference within about 1 SE is inconclusive; "not detected" $\neq$ "no effect"; corroborated, not proven

Q2. With a Laplace prior, northness gets 0.08 (90% interval -0.07 to 0.28); Monday's lasso said exactly 0. Which would you report, and why?

- *Land:* the posterior shows the uncertainty a zero hides; for prediction either works, for inference report the interval

If time: Bayes factors (ratio of marginal likelihoods; sensitive to vague priors). RJMCMC and spike-and-slab put model choice inside the sampler (not in Stan; JAGS, NIMBLE)